

USER_MANUAL

User Manual

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Getting Started

What is the Translator System?

The Universal Multi-Format Multi-Language Ebook Translation System is a powerful tool for translating ebooks and text documents between languages using state-of-the-art AI models. It supports multiple file formats, translation providers, and quality verification systems.

Key Features

- **Multi-format Support:** FB2, EPUB, TXT, HTML, PDF, DOCX
- **8 AI Translation Providers:** OpenAI, Anthropic, Zhipu, DeepSeek, Qwen, Gemini, Ollama, LlamaCpp
- **High-Quality Translations:** Context-aware literary translation with cultural adaptation
- **Distributed Processing:** Scalable multi-node processing for large projects
- **Quality Verification:** Built-in translation quality assessment
- **Multiple Output Formats:** Export to FB2, EPUB, PDF, DOCX
- **Serbian Script Support:** Both Cyrillic and Latin script options

System Requirements

- **Operating System:** Windows 10+, macOS 10.15+, Linux (Ubuntu 18.04+)
- **Memory:** Minimum 4GB RAM (8GB recommended)
- **Storage:** 1GB free space
- **Network:** Internet connection for API-based providers
- **Python:** Version 3.8 or higher (for standalone scripts)
- **Go:** Version 1.21 or higher (for compiled version)

Installation

Option 1: Binary Installation (Recommended)

1. Download the latest release for your platform from the [Releases page](#)
2. Extract the archive
3. Add the translator directory to your PATH

```
# macOS/Linux
export PATH=$PATH:/path/to/translator
```

```
# Windows (Command Prompt)
set PATH=%PATH%;C:\path\to\translator
```

```
# Windows (PowerShell)
$env:PATH += ";C:\path\to\translator"
```

Option 2: Go Installation

```
go install github.com/digital-vasic/translator/cmd/cli@latest
```

Option 3: Python Package Installation

```
pip install universal-translator
```

Option 4: Docker Installation

```
docker pull digitalvasic/translator:latest
docker run -p 8080:8080 digitalvasic/translator:latest
```

Configuration

Environment Variables

Set up your API keys as environment variables:

```
# OpenAI (required for GPT translations)
export OPENAI_API_KEY="your-openai-api-key"
```

```
# Anthropic Claude
export ANTHROPIC_API_KEY="your-anthropic-api-key"

# Zhipu AI (GLM-4)
export ZHIPU_API_KEY="your-zhipu-api-key"

# DeepSeek
export DEEPSEEK_API_KEY="your-deepseek-api-key"

# Google AI (Gemini)
export GEMINI_API_KEY="your-gemini-api-key"

# Qwen
export QWEN_API_KEY="your-qwen-api-key"

# Optional: Local model paths
export OLLAMA_BASE_URL="http://localhost:11434"
export LLAMACPP_MODEL_PATH="/path/to/your/model.gguf"
```

Configuration File

Create a configuration file at `~/.translator/config.yaml`:

```
server:
  port: 8080
  host: "localhost"

translation:
  default_provider: "openai"
  default_model: "gpt-4"
  temperature: 0.7
  max_tokens: 2000
  timeout: 30

quality:
  minimum_score: 0.8
  enable_verification: true
  check_grammar: true
  check_style: true

serbian:
  default_script: "cyrillic" # or "latin"
  phonetic_transliteration: true

cache:
  enabled: true
  ttl: 86400 # 24 hours
  max_size: 1000
```

Database Configuration (Optional)

For persistent storage and history:

```
database:
  type: "sqlite" # or "postgresql", "mysql"
  connection: "translator.db"
  max_connections: 10
```

Basic Usage

Quick Start

1. Simple Text Translation:

```
translator translate --text "Hello, world!" --from en --to sr
                        --provider openai
```

2. File Translation:

```
translator translate --file book.fb2 --from ru --to sr --
                    provider deepseek
```

3. Batch Translation:

```
translator batch --input-dir ./books --output-dir ./translated
                --from ru --to sr
```

Interactive Mode

Start the interactive translation mode:

```
translator interactive
```

This will launch a command-line interface where you can:

- Enter text to translate
- Upload files
- Configure options
- View translation history

Command Line Interface

Global Options

All commands support these global options:

- `--config, -c`: Path to configuration file
- `--verbose, -v`: Verbose output
- `--quiet, -q`: Minimal output
- `--help, -h`: Show help
- `--version`: Show version information

translate Command

Translate text or files.

translator translate [options] INPUT

Options:

- `--text, -t`: Text to translate (alternative to file input)
- `--from, -f`: Source language (default: auto-detect)
- `--to, -o`: Target language (required)
- `--provider, -p`: Translation provider (default: openai)
- `--model, -m`: Model to use (provider-specific)
- `--output, -o`: Output file path
- `--format, -F`: Output format (same as input by default)
- `--script, -s`: Serbian script (cyrillic/latin, default: cyrillic)

Examples:

Translate Russian FB2 to Serbian

```
translator translate book_ru.fb2 --from ru --to sr --provider  
deepseek --output book_sr.fb2
```

Translate single text

```
translator translate --text "Hello, world!" --from en --to sr --  
provider openai
```

Translate with specific model

```
translator translate book.fb2 --from ru --to sr --provider  
anthropic --model claude-3-opus
```

Translate to Latin Serbian script

```
translator translate book.fb2 --from ru --to sr --script latin
```

batch Command

Translate multiple files.

translator batch [options] INPUT_DIRECTORY

Options:

- `--output-dir, -o`: Output directory (required)
- `--from, -f`: Source language
- `--to, -t`: Target language (required)
- `--provider, -p`: Translation provider
- `--pattern, -P`: File pattern (default: *.fb2)
- `--parallel, -j`: Number of parallel workers (default: 4)
- `--resume, -r`: Resume interrupted batch job

Examples:

Translate all FB2 files

```
translator batch ./books --output-dir ./translated --from ru --to  
sr
```

Translate specific pattern

```
translator batch ./books --output-dir ./translated --pattern  
"*.epub" --from en --to sr
```

Use 8 parallel workers

```
translator batch ./books --output-dir ./translated --from ru --to  
sr --parallel 8
```

quality Command

Check translation quality.

```
translator quality [options] FILE
```

Options:

- `--original, -O`: Original text/file
- `--translated, -T`: Translated text/file
- `--from, -f`: Source language
- `--to, -t`: Target language
- `--detailed, -d`: Show detailed analysis

Examples:

Check translation quality

```
translator quality --original book_ru.fb2 --translated  
book_sr.fb2 --from ru --to sr
```

Detailed quality analysis

```
translator quality --original book_ru.fb2 --translated  
book_sr.fb2 --from ru --to sr --detailed
```

server Command

Start the web server.

```
translator server [options]
```

Options:

- `--port, -p`: Port number (default: 8080)
- `--host, -h`: Host address (default: localhost)
- `--dev, -d`: Development mode with hot reload

config Command

Manage configuration.

```
translator config [SUBCOMMAND] [options]
```

Subcommands:

- `show`: Show current configuration
- `set KEY VALUE`: Set configuration value
- `init`: Initialize configuration file
- `test`: Test configuration and API keys

Examples:

Show configuration

```
translator config show
```

Set default provider

```
translator config set default_provider deepseek
```

Test configuration

```
translator config test
```

Initialize config file

```
translator config init
```

Web Interface

Starting the Web Server

```
translator server --port 8080
```

Then open your browser to <http://localhost:8080>.

Web Interface Features

1. **Dashboard**: Overview of translation activity and system status
2. **Upload & Translate**: Drag-and-drop file translation
3. **Batch Processing**: Queue and monitor batch jobs
4. **Quality Reports**: View translation quality assessments
5. **History**: Translation history and statistics
6. **Settings**: Configure providers and options

Uploading Files

1. Navigate to the **Translate** page
2. Drag and drop your file or click to browse
3. Select source and target languages
4. Choose translation provider and model
5. Click **Translate**

Batch Processing

1. Go to the **Batch** page
2. Upload multiple files or select a directory

3. Configure translation options
4. Start the batch job
5. Monitor progress in real-time

Translation Methods

1. AI LLM Translation (Recommended)

Provider Options:

OpenAI GPT

- **Models:** GPT-3.5 Turbo, GPT-4, GPT-4 Turbo
- **Strengths:** Excellent general translation, good context understanding
- **Use Case:** General-purpose translation, balanced quality and cost

Anthropic Claude

- **Models:** Claude 3 Sonnet, Claude 3 Opus
- **Strengths:** Literary translation, maintains authorial voice
- **Use Case:** Literary works, novels, poetry

Zhipu AI (GLM-4)

- **Models:** GLM-4
- **Strengths:** Excellent for Russian-Slavic language pairs
- **Use Case:** Russian to Serbian translation

DeepSeek

- **Models:** DeepSeek-V2
- **Strengths:** Cost-effective, good technical translation
- **Use Case:** Large projects, technical documentation

Qwen

- **Models:** Qwen-Max
- **Strengths:** Good for Asian-European language pairs
- **Use Case:** Multilingual projects

Gemini

- **Models:** Gemini Pro
- **Strengths:** Google's latest model, good cultural adaptation
- **Use Case:** Modern content, web content

Ollama (Local)

- **Models:** Llama 3, Mixtral, custom models

- **Strengths:** Free, offline, private
- **Use Case:** Sensitive content, offline work

LlamaCpp (Local)

- **Models:** GGUF format models
- **Strengths:** Custom models, optimized inference
- **Use Case:** Specialized domains, research

2. Dictionary-Based Translation

Features:

- Fast, no API costs
- Predefined word pairs
- Limited to dictionary terms
- Good for basic translations

Use Case:

- Quick reference
- Simple texts
- Learning vocabulary

3. Hybrid Translation

Combine AI translation with dictionary verification for maximum accuracy:

```
translator translate --file book.fb2 --from ru --to sr --provider  
deepseek --verify-dictionary
```

File Formats

Input Formats

FB2 (FictionBook2)

- **Description:** Russian ebook format
- **Features:** Rich metadata, good for fiction
- **Encoding:** UTF-8
- **Best for:** Novels, stories, literary works

EPUB

- **Description:** Standard ebook format
- **Features:** Reflowable content, multi-device support
- **Encoding:** UTF-8
- **Best for:** Modern fiction, non-fiction

TXT

- **Description:** Plain text format
- **Features:** Simple, universal compatibility
- **Encoding:** UTF-8 or detection
- **Best for:** Simple documents, testing

HTML

- **Description:** Web page format
- **Features:** Structured content, styling
- **Encoding:** UTF-8
- **Best for:** Web content, documentation

PDF

- **Description:** Portable Document Format
- **Features:** Fixed layout, universal viewing
- **Encoding:** Text extraction
- **Best for:** Scanned books, fixed-layout content

DOCX

- **Description:** Microsoft Word format
- **Features:** Rich formatting, track changes
- **Encoding:** UTF-8
- **Best for:** Documents, manuscripts

Output Formats

The same formats are supported for output, with additional options:

- **Preserve Formatting:** Keep original layout and styling
- **Metadata Translation:** Translate titles and metadata
- **Script Options:** Choose Cyrillic or Latin for Serbian

Quality Assurance

Automatic Quality Checks

The system performs automatic quality assessment:

1. **Grammar Check:** Verify grammar rules
2. **Style Consistency:** Check for consistent style
3. **Cultural Adaptation:** Verify cultural references
4. **Technical Accuracy:** Check terminology consistency

Quality Scoring

Translations receive a quality score from 0.0 to 1.0:

- **0.9 - 1.0:** Excellent quality, publication-ready
- **0.8 - 0.9:** Good quality, minimal editing needed
- **0.7 - 0.8:** Acceptable, requires review
- **0.6 - 0.7:** Fair, significant editing required
- **Below 0.6:** Poor, re-translation recommended

Manual Review Process

1. **Preview:** Review translated segments
2. **Compare:** Side-by-side original and translation
3. **Edit:** Make corrections and improvements
4. **Verify:** Re-check quality score
5. **Finalize:** Approve for output

Translation Notes

Add notes and comments to translations:

```
translator translate --file book.fb2 --from ru --to sr --add-notes
```

This will add notes for:

- Uncertain translations
- Cultural references
- Terminology choices
- Formatting issues

Troubleshooting

Common Issues

API Key Problems

Error: "Invalid API key" or "Authentication failed"

Solutions:

1. Verify API key is correct
2. Check environment variable spelling
3. Ensure API key has sufficient credits
4. Try refreshing the key

Verification:

```
translator config test
```

Translation Quality Issues

Problem: Translation quality is poor

Solutions:

1. Try a different provider
2. Adjust temperature (lower = more consistent, higher = more creative)
3. Use a larger model
4. Enable quality verification
5. Split large texts into smaller chunks

Example:

```
translator translate --file book.fb2 --from ru --to sr --provider  
openai --model gpt-4 --temperature 0.5 --verify
```

Memory Issues

Problem: Out of memory errors with large files

Solutions:

1. Use chunked processing:

```
translator translate --file large_book.fb2 --from ru --to sr --  
chunk-size 5000
```

2. Increase memory limits
3. Use batch processing for very large files

Network Issues

Problem: Connection timeouts, network errors

Solutions:

1. Increase timeout:

```
translator translate --file book.fb2 --from ru --to sr --  
timeout 120
```

2. Use local provider (Ollama/LlamaCpp)
3. Enable retry logic:

```
translator translate --file book.fb2 --from ru --to sr --retry  
3
```

Format Issues

Problem: Input file not recognized or corrupted

Solutions:

1. Validate file format:

```
translator validate --file book.fb2
```

2. Convert to supported format
3. Check file encoding (should be UTF-8)
4. Repair corrupted files

Debug Mode

Enable verbose logging for troubleshooting:

```
translator translate --file book.fb2 --from ru --to sr --verbose
```

Log Files

Log files are stored at:

- **Linux/macOS:** ~/.translator/logs/
- **Windows:** %APPDATA%/translator/logs/

Support

For additional help:

1. **Documentation:** Check the [API Documentation](#)
2. **Community:** Join our [Discord server](#)
3. **Issues:** Report bugs at [GitHub Issues](#)
4. **Email:** support@translator.digital

Performance Tips

1. Choose the right provider:

- For quality: Claude 3 Opus, GPT-4
- For cost: DeepSeek, Qwen
- For privacy: Ollama, LlamaCpp

2. Optimize batch processing:

- Use appropriate parallel worker count
- Monitor resource usage
- Resume interrupted jobs

3. Cache translations:

- Enable caching to avoid re-translation
- Use consistent prompts
- Save translation memory

4. Memory management:

- Process large files in chunks
- Monitor system resources
- Use appropriate chunk size

Keyboard Shortcuts

In the web interface:

- **Ctrl/Cmd + O**: Open file
 - **Ctrl/Cmd + S**: Save translation
 - **Ctrl/Cmd + Enter**: Start translation
 - **Ctrl/Cmd + P**: Print/Export
 - **F11**: Fullscreen mode
 - **Esc**: Cancel operation
-

Next Steps

- Explore the [API Documentation](#) for programmatic usage
- Check the [Developer Guide](#) for advanced configuration
- Join our community for tips and best practices
- Try the [Example Projects](#)